

# SHARP COEFFICIENTWISE POSITIVITY FOR A MATRIX TURÁN DETERMINANT OF ${}_0F_1$

JOHN SHIELDS

ABSTRACT. Let  $Z(a, \lambda) = {}_0F_1(; a; \lambda)/\Gamma(a) = \sum_{k \geq 0} \lambda^k / (k! \Gamma(a + k))$  for  $a > 0$ . We prove a sharp coefficientwise positivity theorem for a confluent matrix Turán determinant of this reciprocal-gamma normalization. From parameter and Euler derivatives of  $Z$ , we form a one-parameter family of symmetric  $2 \times 2$  matrices  $\mathcal{T}_\kappa(a, \lambda)$  and prove that, for each fixed  $a > 0$ , every positive-degree Maclaurin coefficient of  $\Delta^{(\kappa)} = \det \mathcal{T}_\kappa$  is strictly positive if and only if  $\kappa \geq 1$ . Thus  $\kappa = 1$  is the exact deformation threshold at every parameter value, and  $\Delta^{(\kappa)}(a, \cdot)$  is strictly absolutely monotone for every  $\kappa \geq 1$ . At this endpoint, coefficientwise determinant positivity holds even though the first nonconstant coefficient matrix may be indefinite: a mixed-determinant polarization confines its possible negative contribution to the degree-two coefficient, whose positivity follows from an explicit decomposition.

Via the identity  $I_{a-1}(2\sqrt{\lambda}) = \lambda^{(a-1)/2} Z(a, \lambda)$ , an exact diagonal congruence transfers the endpoint theorem to the modified Bessel function  $I_\nu$ , yielding, for every  $\nu > -1$  and  $z > 0$ , a positive-definite Bessel–Schur matrix coupling order, argument, and mixed order–argument derivatives of  $\log I_\nu$ . For each fixed  $\nu$ , neither the deformation threshold  $\kappa = 1$  nor the argument-independent correction  $4/\psi_1(\nu + 1)$  can be lowered.

## 1. INTRODUCTION

Turán inequalities for modified Bessel functions are closely connected with order monotonicity, logarithmic derivatives, and total positivity; see [1, 2, 4, 25]. A complementary literature proves scalar log-concavity and coefficientwise Turán inequalities for reciprocal-gamma and hypergeometric series [11, 12, 13, 15, 16]. Here we study the normalized series

$$Z(a, \lambda) = \sum_{k \geq 0} \frac{\lambda^k}{k! \Gamma(a + k)}$$

through a coupled  $2 \times 2$  matrix involving parameter and Euler derivatives.

The entries of that matrix,  $\mathcal{T}_\kappa(a, \lambda)$  of (2.5), are normalized logarithmic derivatives of  $Z$  in the parameter direction  $\partial_a$  and the Euler direction  $\lambda \partial_\lambda$ , together with the mixed derivative; a real parameter  $\kappa$  scales the Euler-direction term alone. Its determinant  $\Delta^{(\kappa)} = \det \mathcal{T}_\kappa$  vanishes at  $\lambda = 0$ , and the central result, Theorem 2.2, decides for which  $\kappa$  all of its remaining Maclaurin coefficients are positive: for each fixed  $a > 0$ , every coefficient of positive degree in  $\Delta^{(\kappa)}(a, \cdot)$  is strictly positive if and only if  $\kappa \geq 1$ . Coefficientwise positivity is strictly stronger than positivity of  $\Delta^{(\kappa)}$  itself, and at the endpoint  $\kappa = 1$  it makes  $\Delta = \Delta^{(1)}(a, \cdot)$  strictly absolutely monotone on  $(0, \infty)$ . Through the identity  $I_{a-1}(2\sqrt{\lambda}) = \lambda^{(a-1)/2} Z(a, \lambda)$ , the endpoint case yields Theorem 2.4: for every  $\nu > -1$  and  $z > 0$ ,

$$(-\partial_\nu^2 \log I_\nu) \left( 2(\Theta_z \log I_\nu - \nu) - \Theta_z^2 \log I_\nu + \frac{4}{\psi_1(\nu + 1)} \right) > (1 + \partial_\nu \Theta_z \log I_\nu)^2,$$

where  $\Theta_z = z \partial_z$  and  $\psi_1$  is the trigamma function. The matrix behind this inequality, the Bessel–Schur matrix of Corollary 2.5, couples the concavity of  $\log I_\nu$  in the order, the argument-direction

---

*Date:* July 28, 2026.

*2020 Mathematics Subject Classification.* Primary 33C10; Secondary 15A45, 26A48, 26D15.

*Key words and phrases.* coefficientwise positivity, Turán inequality, modified Bessel function, reciprocal-gamma series, hypergeometric function, order derivative, absolute monotonicity.

Independent researcher, Ginza 2-15-2, KR Ginza II 5F, Chuo-ku, Tokyo, Japan 104-0061. Corresponding author: math@johnnyshields.com. ORCID: 0009-0004-3882-4571. This work is archived at doi:10.5281/zenodo.21438795.

expression above, and the mixed order–argument sensitivity;  $\Delta$  is the corresponding differential Schur determinant. By a *matrix Turán determinant* we mean this confluent Turánian assembled from derivatives, rather than a determinant of parameter-shifted evaluations (compared in Section 9).

A single threshold in  $\kappa$  serves every  $a > 0$  and is exact at each  $a$  separately, the two failures being detected respectively in degree one as  $a \downarrow 0$  and at large argument (Corollary 7.2). On the Bessel side, Proposition 2.6 identifies  $4/\psi_1(\nu + 1)$  as the least argument-independent correction: no smaller constant keeps the displayed inequality true for all  $z > 0$ . Both sharpness statements hold on the positive-series domain  $a > 0$  (equivalently  $\nu > -1$ ), the range in which every coefficient of  $Z$  is positive.

Section 3 opens with an asymmetric Chu–Vandermonde convolution, which yields exact formulas for the coefficient matrices of  $\mathcal{T}_\kappa$ . These formulas expose the structure of the sharp endpoint  $\kappa = 1$ : the convolution identity singles out  $m = 1$  as the potentially exceptional positive degree.

In Section 4, explicit  $\ell^2$  Gram representations prove that every coefficient matrix of degree  $m \geq 2$  is positive definite, whereas  $M_1$  can be indefinite for small  $a > 0$ . Section 5 then passes, by polarization, from positivity of the individual coefficient matrices to positivity of the determinant coefficients. The potentially indefinite matrix  $M_1$  can contribute negatively only through its self-pairing, which occurs solely in the coefficient  $\Delta_2$ ; an independent explicit decomposition proves that this exceptional coefficient is nevertheless strictly positive. This completes the endpoint theorem. Comparison of  $\mathcal{T}_\kappa$  with  $\mathcal{T}_1$  then gives sufficiency for  $\kappa \geq 1$  and the sharpness of the common threshold.

Finally, in Sections 6 and 7, an explicit diagonal congruence transports the endpoint theorem to a positive-definite Bessel–Schur matrix and a mixed order–argument inequality for every  $\nu > -1$ . The same reduction yields the two sharpness statements of Proposition 2.6 and, in the large-argument limit, the necessity at fixed  $a$ . Section 8 examines the boundary of the positive-series domain: although the leading small-argument term remains positive away from the poles, the continued determinant already loses coefficientwise positivity on  $-3 < \nu < -2$  (Proposition 8.2).

## 2. MAIN RESULTS

Throughout,  $a > 0$ ,  $\lambda \geq 0$ , and  $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ . We use

$$(2.1) \quad Z(a, \lambda) = \sum_{k=0}^{\infty} \frac{\lambda^k}{k! \Gamma(a+k)} = \frac{{}_0F_1(; a; \lambda)}{\Gamma(a)}.$$

The series in (2.1) and all fixed-order  $a$ - and Euler derivatives used below converge locally uniformly, and extend to entire functions of  $(a, \lambda) \in \mathbb{C}^2$  (Lemma 8.1); the standing assumptions  $a > 0$  and  $\lambda \geq 0$  are in force until Section 8, where they are relaxed. Put  $\Theta_\lambda = \lambda \partial_\lambda$  and  $g = \psi_1(a)$ , where  $\psi = \psi^{(0)}$  and  $\psi_1 = \psi^{(1)}$  are the digamma and trigamma functions. In the Bessel variable we use the distinct notation  $\Theta_z = z \partial_z$ . Subscripts denote application of the indicated operator, so  $Z_a = \partial_a Z$ ,  $Z_{\Theta_\lambda} = \Theta_\lambda Z$ , and  $Z_{a\Theta_\lambda} = \partial_a \Theta_\lambda Z$ . For  $\kappa \in \mathbb{R}$ , define

$$(2.2) \quad A = Z_a^2 - Z Z_{aa},$$

$$(2.3) \quad B = Z^2 + Z Z_{a\Theta_\lambda} - Z_a Z_{\Theta_\lambda},$$

$$(2.4) \quad C_\kappa = Z^2 + g (\kappa Z Z_{\Theta_\lambda} - Z Z_{\Theta_\lambda \Theta_\lambda} + Z_{\Theta_\lambda}^2),$$

$$(2.5) \quad \mathcal{T}_\kappa = \begin{pmatrix} A & \sqrt{g} B \\ \sqrt{g} B & C_\kappa \end{pmatrix}, \quad \Delta^{(\kappa)} = \det \mathcal{T}_\kappa = AC_\kappa - gB^2.$$

At  $\kappa = 1$  write  $C = C_1$ ,  $\mathcal{T} = \mathcal{T}_1$ , and  $\Delta = \Delta^{(1)}$ .

**Remark 2.1.** Writing  $L = \log Z$ , direct differentiation gives

$$\frac{A}{Z^2} = -L_{aa}, \quad \frac{B}{Z^2} = 1 + L_{a\Theta_\lambda}, \quad \frac{C_\kappa}{gZ^2} = \frac{1}{g} + \kappa L_{\Theta_\lambda} - L_{\Theta_\lambda \Theta_\lambda}.$$

Thus the entries are normalized logarithmic derivatives of  $Z$  in the parameter and Euler directions — a curvature in the parameter direction, a shifted mixed derivative, and a combination of first and second Euler derivatives — with  $\kappa$  deforming the argument-direction term. At the endpoint  $\kappa = 1$ , the normalization is chosen so that the identity (6.1) and a diagonal congruence carry  $\mathcal{T}$  to the order–argument Bessel–Schur matrix of Corollary 2.5; see Section 6.

A function on  $(0, \infty)$  is *strictly absolutely monotone* if it and all of its derivatives are strictly positive there.

**Theorem 2.2** (Sharp coefficientwise classification). *For each  $a > 0$ ,*

$$(2.6) \quad \Delta^{(\kappa)}(a, \lambda) = \sum_{n \geq 1} \Delta_n^{(\kappa)}(a) \lambda^n, \quad \Delta_n^{(\kappa)}(a) > 0 \quad \text{for every } n \geq 1,$$

*holds if and only if  $\kappa \geq 1$ . The threshold is therefore exact at every parameter value, and a single threshold serves every  $a > 0$ . For  $\kappa < 1$  the linear coefficient is negative for all sufficiently small  $a > 0$ , while at a fixed  $a > 0$  the first negative coefficient can occur in arbitrarily high degree as  $\kappa \uparrow 1$  (Remark 7.3); the necessity at fixed  $a$  is proved in Corollary 7.2. Consequently, for every  $\kappa \geq 1$ ,  $\Delta^{(\kappa)}(a, \cdot)$  is strictly absolutely monotone on  $(0, \infty)$  and  $\mathcal{T}_\kappa(a, \lambda) \succ 0$  for  $a, \lambda > 0$ .*

That a single threshold serves every  $a > 0$  does not assert a positive lower bound uniform in  $a$ .

The series in (2.6) starts at degree one because

$$A(a, 0) = \frac{g}{\Gamma(a)^2}, \quad B(a, 0) = C_\kappa(a, 0) = \frac{1}{\Gamma(a)^2},$$

so  $\mathcal{T}_\kappa(a, 0) = \Gamma(a)^{-2} \begin{pmatrix} g & \sqrt{g} \\ \sqrt{g} & 1 \end{pmatrix}$  has rank one and  $\Delta^{(\kappa)}(a, 0) = 0$ .

**Remark 2.3.** The rank-one constant term is shared by the entire family  $\mathcal{T}_\kappa$ , so the sign of the first nonzero determinant coefficient detects the sharp threshold in  $\kappa$  uniformly in  $a$ , although positivity in every degree for  $\kappa \geq 1$  requires the full coefficientwise argument.

Under the correspondence  $a = \nu + 1$ ,  $z = 2\sqrt{\lambda}$  of the identity (6.1), order and argument derivatives of  $I_\nu$  correspond to  $a$ - and Euler derivatives of  $Z$ , which exist by the locally uniform convergence recorded after (2.1) and the entirety in  $(a, \lambda)$  of Lemma 8.1. For  $\nu > -1$  and  $z > 0$  the positive coefficients of  $Z(a, \lambda)$  show that  $I_\nu(z) > 0$ , so  $\log I_\nu$  is real and the following logarithmic derivatives are well defined:

$$(2.7) \quad G_\nu = -\partial_\nu^2 \log I_\nu,$$

$$(2.8) \quad P_\nu = \partial_\nu(\Theta_z \log I_\nu),$$

$$(2.9) \quad H_\nu = 2(\Theta_z \log I_\nu - \nu) - \Theta_z^2 \log I_\nu,$$

and

$$(2.10) \quad D_\nu = G_\nu \left( H_\nu + \frac{4}{\psi_1(\nu + 1)} \right) - (1 + P_\nu)^2.$$

The shifts in  $1 + P_\nu$  and  $4/\psi_1(\nu + 1)$  are exactly those produced by the diagonal congruence (6.5). With  $L = \log Z$  as in Remark 2.1,  $P_\nu = 1 + 2L_{a\Theta_\lambda}$ , so  $1 + P_\nu = 2(1 + L_{a\Theta_\lambda})$ ; as  $z \downarrow 0$  one has  $L_{a\Theta_\lambda} \rightarrow 0$ , hence  $P_\nu \rightarrow 1$  and  $1 + P_\nu \rightarrow 2$ . For  $\kappa \in \mathbb{R}$ , define the deformation

$$(2.11) \quad H_\nu^{(\kappa)} = 2\kappa(\Theta_z \log I_\nu - \nu) - \Theta_z^2 \log I_\nu,$$

$$(2.12) \quad D_\nu^{(\kappa)} = G_\nu \left( H_\nu^{(\kappa)} + \frac{4}{\psi_1(\nu + 1)} \right) - (1 + P_\nu)^2.$$

At  $\kappa = 1$  these reduce to  $H_\nu$  and  $D_\nu$ . We call this the *fixed-correction deformation* because the argument-independent correction  $4/\psi_1(\nu + 1)$  is held fixed while the argument-direction term is deformed.

**Theorem 2.4** (Mixed Bessel–Schur inequality). *For  $\nu > -1$  and  $z > 0$ ,  $D_\nu(z) > 0$ , equivalently*

$$(2.13) \quad G_\nu(z) \left( H_\nu(z) + \frac{4}{\psi_1(\nu+1)} \right) > (1 + P_\nu(z))^2.$$

**Corollary 2.5** (Positive Bessel–Schur matrix). *For  $\nu > -1$  and  $z > 0$ ,*

$$\begin{pmatrix} G_\nu(z) & 1 + P_\nu(z) \\ 1 + P_\nu(z) & H_\nu(z) + 4/\psi_1(\nu+1) \end{pmatrix} \succ 0.$$

*As  $z \downarrow 0$ , the matrix-valued function admits a continuous extension whose value is the rank-one matrix  $\begin{pmatrix} g & 2 \\ 2 & 4/g \end{pmatrix}$ , where  $g = \psi_1(\nu+1)$ .*

**Proposition 2.6** (Sharpness of the Bessel correction and deformation). *For each fixed  $\nu > -1$ ,  $4/\psi_1(\nu+1)$  is the least real constant  $R$  such that*

$$G_\nu(z)(H_\nu(z) + R) > (1 + P_\nu(z))^2 \quad \text{for every } z > 0.$$

*Moreover, for the fixed-correction deformation  $D_\nu^{(\kappa)}$  defined in (2.12) and each fixed  $\nu > -1$ ,*

$$D_\nu^{(\kappa)}(z) > 0 \quad \text{for every } z > 0$$

*holds if and only if  $\kappa \geq 1$ .*

Sufficiency and the sharpness of the common threshold in Theorem 2.2 are proved in Section 5, and the necessity at each fixed  $a > 0$  is completed in Corollary 7.2; Theorem 2.4, corollary 2.5, and proposition 2.6 are proved in Section 7.

### 3. RECIPROCAL-GAMMA CONVOLUTION AND COEFFICIENT FORMULAS

The coefficient calculation rests on a two-parameter Chu–Vandermonde identity.

**Lemma 3.1** (Asymmetric reciprocal-gamma convolution). *Let  $m \in \mathbb{N}_0$  and let  $\alpha, \beta > 0$ . Then*

$$(3.1) \quad \sum_{i=0}^m \frac{1}{i!(m-i)!\Gamma(\alpha+i)\Gamma(\beta+m-i)} = \frac{(\alpha+\beta+m-1)_m}{m!\Gamma(\alpha+m)\Gamma(\beta+m)}.$$

*Both sides are entire in  $(\alpha, \beta)$ .*

*Proof.* Multiply the left-hand side by  $m!\Gamma(\alpha+m)\Gamma(\beta+m)$ . Using

$$\frac{\Gamma(\alpha+m)}{\Gamma(\alpha+i)} = (\alpha+i)_{m-i} = \frac{(\alpha)_m}{(\alpha)_i}$$

and

$$\frac{\Gamma(\beta+m)}{\Gamma(\beta+m-i)} = (\beta+m-i)_i = (-1)^i(1-\beta-m)_i,$$

the resulting sum is

$$(\alpha)_m {}_2F_1(-m, 1-\beta-m; \alpha; 1).$$

Chu–Vandermonde gives  $(\alpha+\beta+m-1)_m$ . Both sides of (3.1) are entire in  $(\alpha, \beta)$ , being finite sums and products of the entire function  $1/\Gamma$  and a polynomial, so agreement on  $\{\alpha, \beta > 0\}$  extends the identity to  $\mathbb{C}^2$  by the identity theorem in each variable separately.  $\square$

Set

$$S_m = S_m(a) = \frac{(2a+m-1)_m}{m!\Gamma(a+m)^2} > 0.$$

Taking  $\alpha = \beta = a$  in Lemma 3.1 gives

$$[\lambda^m]Z(a, \lambda)^2 = S_m,$$

where  $[\lambda^m]$  extracts the coefficient of  $\lambda^m$ . Unsubscripted  $\alpha, \beta$  always denote the parameters of Lemma 3.1; the subscripted families of Theorem 3.2 below are coefficient sequences.

**Theorem 3.2.** *There are expansions*

$$(3.2) \quad A = \sum_{m \geq 0} S_m \alpha_m \lambda^m, \quad B = \sum_{m \geq 0} S_m \beta_m \lambda^m, \quad C_\kappa = \sum_{m \geq 0} S_m \gamma_m^{(\kappa)} \lambda^m,$$

where

$$(3.3) \quad \alpha_m = \psi_1(a + m),$$

$$(3.4) \quad \beta_0 = 1, \quad \beta_m = \frac{2a + m - 2}{2(a + m - 1)} \quad (m \geq 1),$$

and

$$(3.5) \quad \gamma_m^{(\kappa)} = 1 + g c_m^{(\kappa)},$$

with

$$(3.6) \quad c_0^{(\kappa)} = 0, \quad c_1^{(\kappa)} = \frac{\kappa - 1}{2},$$

and, for  $m \geq 2$ ,

$$(3.7) \quad c_m^{(\kappa)} = \frac{\kappa m}{2} - \frac{m(2a + m - 2)}{2(2a + 2m - 3)} = \frac{m[(\kappa - 1)(2a + 2m - 3) + m - 1]}{2(2a + 2m - 3)}.$$

At the sharp endpoint  $\kappa = 1$ ,

$$c_0 = c_1 = 0, \quad c_m = \frac{m(m - 1)}{2(2a + 2m - 3)} \quad (m \geq 2).$$

Consequently,

$$(3.8) \quad \mathcal{T}(a, \lambda) = \sum_{m \geq 0} S_m M_m \lambda^m, \quad M_m = \begin{pmatrix} \alpha_m & \sqrt{g} \beta_m \\ \sqrt{g} \beta_m & 1 + g c_m \end{pmatrix}.$$

*Proof.* For the first coefficient, define

$$F_m(\delta) = \sum_{i=0}^m \frac{1}{i!(m-i)!\Gamma(a+\delta+i)\Gamma(a-\delta+m-i)}.$$

By Lemma 3.1,

$$(3.9) \quad F_m(\delta) = \frac{(2a + m - 1)_m}{m!\Gamma(a + m + \delta)\Gamma(a + m - \delta)}.$$

On the one hand,

$$(3.10) \quad \partial_\delta^2 [Z(a + \delta, \lambda)Z(a - \delta, \lambda)]|_{\delta=0} = 2ZZ_{aa} - 2Z_a^2 = -2A.$$

On the other hand,  $F'_m(0) = 0$  and the logarithmic second derivative of (3.9) is  $-2\psi_1(a + m)$  at  $\delta = 0$ , so

$$(3.11) \quad F''_m(0) = -2S_m \psi_1(a + m).$$

Comparing the coefficient of  $\lambda^m$  in (3.10) with (3.11) gives  $[\lambda^m]A = S_m \psi_1(a + m)$ , proving (3.3).

For  $m = 0$ , one has  $[\lambda^0]B = Z(a, 0)^2 = S_0$ , so  $\beta_0 = 1$ . Assume henceforth that  $m \geq 1$ . Let  $l$  be the index of the first factor in  $Z(a + \delta, \lambda)Z(a - \delta, \lambda) = \sum_m F_m(\delta)\lambda^m$ , so that  $[\lambda^m]$  weights  $i$  by

$$w_i(\delta) = [i!(m-i)!\Gamma(a+\delta+i)\Gamma(a-\delta+m-i)]^{-1}.$$

For real  $\delta$  with  $|\delta| < a$ , write  $\mathbb{E}_\delta$  for expectation under the probability weights  $w_i(\delta)/F_m(\delta)$ , and put  $\mathbb{E} = \mathbb{E}_0$ ,  $D = m - 2l$ , and  $J = m - l$ . Applying Lemma 3.1 to  $\sum_i i w_i(\delta)$ , with  $(\alpha, \beta) = (a + 1 + \delta, a - \delta)$  and  $m - 1$  in place of  $m$ , gives

$$\sum_{i=0}^m i w_i(\delta) = \frac{(2a + m - 1)_{m-1}}{(m-1)!\Gamma(a + m + \delta)\Gamma(a + m - 1 - \delta)}.$$

Dividing by (3.9) yields

$$\mathbb{E}_\delta \mathbf{l} = \frac{m(a+m-1-\delta)}{2(a+m-1)},$$

whence

$$(3.12) \quad \mathbb{E}_\delta \mathbf{D} = \frac{m\delta}{a+m-1}.$$

The score  $\partial_\delta \log w_i(\delta)|_{\delta=0}$  is

$$X = \psi(a+m-1) - \psi(a+1),$$

so differentiating (3.12) at  $\delta = 0$  gives  $\mathbb{E}(\mathbf{D}X) = \partial_\delta \mathbb{E}_\delta \mathbf{D}|_{\delta=0}$ , that is

$$(3.13) \quad \mathbb{E}(\mathbf{D}X) = \frac{m}{a+m-1}.$$

Direct coefficient extraction gives

$$\begin{aligned} & [\lambda^m](ZZ_{a\Theta_\lambda} - Z_a Z_{\Theta_\lambda}) \\ &= \sum_{i=0}^m \frac{-(m-i)\psi(a+m-i) + (m-i)\psi(a+i)}{i!(m-i)!\Gamma(a+i)\Gamma(a+m-i)} = -S_m \mathbb{E}(\mathbf{J}X). \end{aligned}$$

Together with  $[\lambda^m]Z^2 = S_m$ , this yields

$$\frac{[\lambda^m]B}{S_m} = 1 - \mathbb{E}(\mathbf{J}X) = 1 - \frac{1}{2}\mathbb{E}(\mathbf{D}X),$$

because  $\mathbf{J} = (m + \mathbf{D})/2$  and  $\mathbb{E}X = 0$  by symmetry. This proves (3.4).

For  $C_\kappa$ , the Euler-derivative terms contribute

$$\kappa \mathbb{E}\mathbf{J} - \mathbb{E}\mathbf{J}^2 + \mathbb{E}(\mathbf{I}\mathbf{J}).$$

Since  $\mathbf{l} = (m - \mathbf{D})/2$ ,  $\mathbf{J} = (m + \mathbf{D})/2$ , and  $\mathbb{E}\mathbf{D} = 0$  by the symmetry  $\mathbf{l} \leftrightarrow m - \mathbf{l}$ ,

$$(3.14) \quad \kappa \mathbb{E}\mathbf{J} - \mathbb{E}\mathbf{J}^2 + \mathbb{E}(\mathbf{I}\mathbf{J}) = \frac{\kappa m}{2} - \frac{1}{2}\mathbb{E}\mathbf{D}^2.$$

For  $m \geq 2$ , cancelling the factors  $i(m-i)$  and shifting  $j = i-1$  gives

$$\begin{aligned} S_m(a)\mathbb{E}[\mathbf{l}(m-\mathbf{l})] &= \sum_{i=1}^{m-1} \frac{1}{(i-1)!(m-i-1)!\Gamma(a+i)\Gamma(a+m-i)} \\ &= S_{m-2}(a+1). \end{aligned}$$

Therefore

$$\mathbb{E}[\mathbf{l}(m-\mathbf{l})] = \frac{S_{m-2}(a+1)}{S_m(a)} = \frac{m(m-1)(a+m-1)}{2(2a+2m-3)},$$

so

$$\mathbb{E}\mathbf{D}^2 = \frac{m(2a+m-2)}{2a+2m-3}.$$

Substitution gives (3.7); the cases  $m = 0, 1$  are immediate.  $\square$

**Remark 3.3.** Theorem 3.2 does not assert  $M_m \succeq 0$  for every  $m$ :  $M_1$  can be indefinite for small  $a > 0$  (see Section 4). For general  $\kappa$ , write  $M_m^{(\kappa)}$  for the coefficient matrix of  $\mathcal{T}_\kappa$ , obtained by replacing  $1 + gc_m$  with  $1 + gc_m^{(\kappa)}$ .

**Remark 3.4.** The weights  $w_i(0)$  of the proof of Theorem 3.2, normalized by  $F_m(0) = S_m$ , form a symmetric finite law on  $0 \leq i \leq m$ , under which  $\mathbf{l}$  is the random index;  $\mathbb{E}$ ,  $\mathbf{D}$ , and  $X$  are as defined there, and

$$\sigma = \psi_1(a+1) + \psi_1(a+m-1).$$

The three normalized coefficient entries may be written as

$$\alpha_m = \frac{1}{2}\mathbb{E}(\sigma - X^2), \quad \beta_m = 1 - \frac{1}{2}\mathbb{E}(\mathbf{D}X), \quad c_m^{(\kappa)} = \frac{\kappa m}{2} - \frac{1}{2}\mathbb{E}\mathbf{D}^2.$$

The first identity follows by writing

$$\frac{F_m''(0)}{F_m(0)} = \mathbb{E}(X^2 - \sigma) = -2\psi_1(a + m),$$

while the other two are the expectation identities obtained in (3.13) and (3.14). Thus the asymmetric Chu–Vandermonde calculation evaluates, respectively, a curvature term, an imbalance–score covariance, and an imbalance variance.

#### 4. GRAM STRUCTURE AND THE EXCEPTIONAL MATRIX $M_1$

We require elementary two-sided trigamma estimates. The series  $\psi_1(y) = \sum_{r \geq 0} (y + r)^{-2}$  is standard [20, eq. 5.15.1]; inserting  $(y + r)^{-2} = \int_0^\infty \tau e^{-(y+r)\tau} d\tau$  and summing  $\sum_{r \geq 0} e^{-r\tau} = (1 - e^{-\tau})^{-1}$  under the integral gives the integral representation

$$(4.1) \quad \psi_1(y) = \int_0^\infty e^{-y\tau} \frac{\tau}{1 - e^{-\tau}} d\tau, \quad y > 0.$$

**Lemma 4.1.** *For every  $y > 0$ ,*

$$(4.2) \quad \psi_1(y) > \frac{1}{y} + \frac{1}{2y^2} > \frac{1}{y}.$$

*For  $y > 1/2$ ,*

$$(4.3) \quad \psi_1(y) < \frac{1}{y - 1/2}.$$

*Consequently,*

$$(4.4) \quad \frac{1}{\psi_1(a)} > a - \frac{1}{2} \quad (a > 0),$$

*where the assertion is immediate for  $a \leq 1/2$ .*

*Proof.* For  $h(s) = (y + s)^{-2}$ , strict convexity on every interval  $[r, r + 1]$  gives the strict trapezoidal inequality

$$\int_r^{r+1} h(s) ds < \frac{h(r) + h(r + 1)}{2}.$$

Summing and passing to the limit yields

$$\psi_1(y) = \sum_{r \geq 0} h(r) > \int_0^\infty h(s) ds + \frac{1}{2}h(0) = \frac{1}{y} + \frac{1}{2y^2},$$

which proves (4.2). For the upper bound, the elementary inequality  $\tau < 2 \sinh(\tau/2)$  for  $\tau > 0$  is equivalent to  $\tau/(1 - e^{-\tau}) < e^{\tau/2}$ . Substitution in (4.1) gives (4.3). Taking reciprocals proves (4.4) when  $a > 1/2$ .  $\square$

Fix  $m \geq 1$  and write

$$x = a + m, \quad q = a + \frac{m}{2} - 1.$$

In  $\ell^2(\mathbb{N}_0)$  define

$$u_r^{(m)} = \frac{1}{x + r}, \quad v_r^{(m)} = \frac{q}{x + r - 1}, \quad r \geq 0.$$

Then

$$(4.5) \quad \|u^{(m)}\|^2 = \psi_1(x) = \alpha_m,$$

$$(4.6) \quad \langle u^{(m)}, v^{(m)} \rangle = q \sum_{r \geq 0} \frac{1}{(x + r - 1)(x + r)} = \frac{q}{x - 1} = \beta_m,$$

and

$$(4.7) \quad \|v^{(m)}\|^2 = q^2 \psi_1(x - 1).$$

Define

$$(4.8) \quad \rho_m(a) = \frac{1}{g} + c_m - q^2 \psi_1(x-1).$$

For vectors  $\xi, \eta$  in a Hilbert space,  $\text{Gram}(\xi, \eta)$  denotes the symmetric  $2 \times 2$  matrix with entries  $\langle \xi, \xi \rangle, \langle \xi, \eta \rangle, \langle \eta, \eta \rangle$ . Write  $N_m = \text{diag}(1, g^{-1/2}) M_m \text{diag}(1, g^{-1/2})$  for the normalized coefficient matrix.

**Theorem 4.2.** *If either  $m \geq 2$  and  $a > 0$ , or  $m = 1$  and  $a \geq 1/2$ , then  $\rho_m(a) > 0$ . In  $\ell^2(\mathbb{N}_0) \oplus \mathbb{R}$ , put*

$$\xi_m = (u^{(m)}, 0), \quad \eta_m = (v^{(m)}, \sqrt{\rho_m(a)}).$$

Then

$$(4.9) \quad N_m = \text{Gram}(\xi_m, \eta_m),$$

$$(4.10) \quad M_m = \text{Gram}(\xi_m, \sqrt{g} \eta_m).$$

In particular,  $M_m$  is positive definite in these ranges.

*Proof.* Assume first that  $m \geq 2$ . Then  $x - 1 > 1/2$ , so Lemma 4.1 gives

$$\rho_m(a) > a - \frac{1}{2} + c_m - \frac{q^2}{x - 3/2}.$$

The right-hand side simplifies exactly to

$$a - \frac{1}{2} + \frac{m(m-1)}{2(2a+2m-3)} - \frac{(a+m/2-1)^2}{a+m-3/2} = \frac{m-1}{2(2a+2m-3)} > 0.$$

For  $m = 1$ ,  $q = a - 1/2$  and  $c_1 = 0$ . If  $a > 1/2$ , then

$$q^2 \psi_1(a) < q < \frac{1}{\psi_1(a)},$$

by (4.3) and (4.4). Hence  $\rho_1(a) > 0$ . At  $a = 1/2$ ,  $q = 0$  and the conclusion is immediate. The Gram identities now follow from (4.5)–(4.8). In a vanishing combination  $\mu \xi_m + \omega \sqrt{g} \eta_m = 0$  the last coordinate gives  $\omega \sqrt{g \rho_m(a)} = 0$ , hence  $\omega = 0$ , and then  $\|u^{(m)}\|^2 = \psi_1(x) > 0$  forces  $\mu = 0$ . The two Gram vectors of (4.10) are therefore linearly independent, so  $M_m \succ 0$ .  $\square$

The constant matrix is

$$M_0 = \begin{pmatrix} g & \sqrt{g} \\ \sqrt{g} & 1 \end{pmatrix} = \begin{pmatrix} \sqrt{g} \\ 1 \end{pmatrix} \begin{pmatrix} \sqrt{g} & 1 \end{pmatrix},$$

a rank-one positive-semidefinite matrix. With  $\psi_1(a+1) = g - a^{-2}$  and  $\beta_1 = (2a-1)/(2a)$ , the remaining coefficient matrix  $M_1$  has determinant

$$(4.11) \quad \det M_1 = \psi_1(a+1) - g\beta_1^2 = \frac{(4a-1)g-4}{4a^2}.$$

**Lemma 4.3.** *There is  $a_* = 0.3690738484 \dots \in (0, 1/2)$  such that  $M_1$  is indefinite for  $0 < a < a_*$ , positive semidefinite of rank one at  $a = a_*$ , and positive definite for  $a > a_*$ .*

*Proof.* Set  $f(a) = (4a-1)\psi_1(a) - 4$ , so that  $\det M_1$  has the sign of  $f(a)$  by (4.11). Since  $g = \psi_1(a) = \sum_{r \geq 0} (a+r)^{-2}$ , for  $0 < a < 1/2$

$$f'(a) = 2 \sum_{r \geq 0} \frac{2r+1-2a}{(a+r)^3} > 0,$$

while  $f(a) \rightarrow -\infty$  as  $a \downarrow 0$  and  $f(1/2) = \psi_1(1/2) - 4 = \pi^2/2 - 4 > 0$ . Hence  $f$  has a unique zero  $a_* \in (0, 1/2)$ . The trichotomy follows with Theorem 4.2 for  $a \geq 1/2$  and the positive diagonal of  $M_1$ .  $\square$

Among the coefficient matrices only  $M_1$  can be indefinite at the endpoint  $\kappa = 1$ , so the determinant positivity of Theorem 2.2 cannot follow from pointwise semidefiniteness of the  $M_m$  alone; see Figure 1.

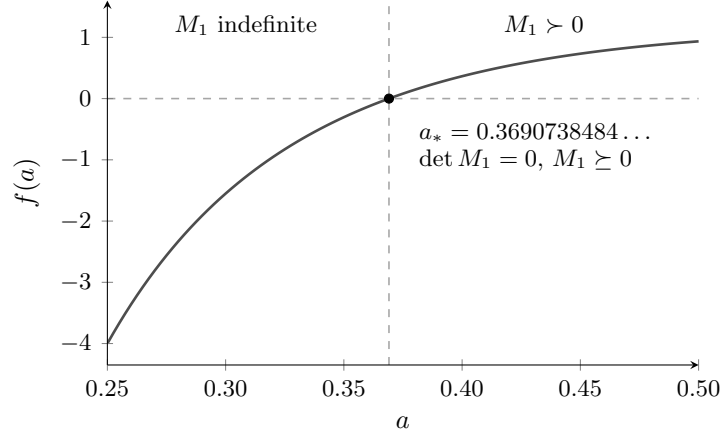


FIGURE 1. By (4.11),  $f(a) = (4a-1)\psi_1(a)-4 = 4a^2 \det M_1$ . The plotted window is fixed by the exact values  $f(1/4) = -4$  and  $f(1/2) = \pi^2/2 - 4$ . Strict increase on  $(0, 1/2)$  makes the zero  $a_* = 0.3690738484\dots$  unique;  $M_1$  is indefinite for  $a < a_*$ , singular positive semidefinite at  $a = a_*$ , and positive definite for  $a > a_*$  (Lemma 4.3). Nevertheless, at  $\kappa = 1$ , every determinant coefficient  $\Delta_n(a)$ ,  $n \geq 1$ , is strictly positive for all  $a > 0$  (Theorem 2.2).

## 5. MIXED DETERMINANTS AND COEFFICIENTWISE POSITIVITY

For symmetric  $2 \times 2$  matrices  $X = (X_{ij})$  and  $Y = (Y_{ij})$ , define

$$\text{MD}(X, Y) = X_{11}Y_{22} + X_{22}Y_{11} - 2X_{12}Y_{12}.$$

This is the polarization of the determinant.

**Lemma 5.1.** *If  $X, Y \succeq 0$ , then  $\text{MD}(X, Y) \geq 0$ . If in addition  $X \neq 0$  and  $Y \succ 0$ , then  $\text{MD}(X, Y) > 0$ .*

*Proof.* Write  $X = \sum_r x_r x_r^T$  and  $Y = \sum_s y_s y_s^T$ . Then

$$\text{MD}(X, Y) = \sum_{r,s} (x_{r,1}y_{s,2} - x_{r,2}y_{s,1})^2.$$

If this sum vanished and  $X \neq 0$ , fix a nonzero Gram vector  $x_r$ . Every Gram vector  $y_s$  would then be parallel to  $x_r$ , forcing  $Y$  to have rank at most one, contrary to  $Y \succ 0$ .  $\square$

Expanding the determinant of (3.8) gives

$$(5.1) \quad \Delta_n = \frac{1}{2} \sum_{k=0}^n S_k S_{n-k} \text{MD}(M_k, M_{n-k}).$$

By Theorem 3.2 and Remark 3.3,  $\mathcal{T}_\kappa(a, \lambda) = \sum_{m \geq 0} S_m M_m^{(\kappa)} \lambda^m$ , so (5.1) holds for  $\Delta_n^{(\kappa)}$  with each  $M_m$  replaced by  $M_m^{(\kappa)}$ .

**Lemma 5.2.** *For every  $a > 0$  and every  $n \geq 1$  with  $n \neq 2$ , one has  $\Delta_n(a) > 0$ .*

*Proof.* The linear coefficient is strictly positive: (4.2) gives  $\psi_1(a) > 1/a$ , so  $ag > 1$  and

$$\text{MD}(M_0, M_1) = \frac{ag - 1}{a^2} > 0,$$

whence

$$(5.2) \quad \Delta_1 = S_0 S_1 \text{MD}(M_0, M_1) = \frac{2(ag - 1)}{a^3 \Gamma(a)^4} > 0.$$

For  $m \geq 2$  the mixed term satisfies

$$(5.3) \quad \text{MD}(M_1, M_m) = \alpha_1(1 + gc_m) + \alpha_m - 2g\beta_1\beta_m > 0.$$

If  $a \geq 1/2$ , then  $M_1, M_m \succ 0$  by Theorem 4.2 and the strict case of Lemma 5.1 applies; if  $0 < a < 1/2$ , then  $\beta_1 < 0 < \beta_m$  by (3.4), and since every diagonal entry is positive the right-hand side is strictly positive. For  $n \geq 3$  the self-pair  $(1, 1)$  does not occur. Classify the pairs  $(k, n - k)$  in (5.1): if neither index is 1, both matrices in the pair are positive semidefinite— $M_0$  is rank-one positive semidefinite and  $M_m \succ 0$  for  $m \geq 2$  by Theorem 4.2—so the nonstrict case of Lemma 5.1 gives a nonnegative summand; if one index is 1, the other is at least 2 and (5.3) applies. Every summand is therefore nonnegative, and the pair  $(M_0, M_n)$  is strictly positive because  $M_0 \neq 0$  and  $M_n \succ 0$ . Hence  $\Delta_n > 0$ .  $\square$

The self-pairing  $\text{MD}(M_1, M_1) = 2 \det M_1$  occurs solely in  $\Delta_2$ , so the indefiniteness established in Lemma 4.3 can contribute negatively only there.

### 5.1. The exceptional determinant coefficient $\Delta_2$ .

**Lemma 5.3.** *For every  $a > 0$ , one has  $\Delta_2(a) > 0$ . More precisely, let*

$$t = \psi_1(a + 1), \quad \mathcal{R}_a = t - \frac{1}{a + 1}.$$

Then

$$(5.4) \quad \Delta_2 = \frac{Q_*(a, t)}{2a^6(a + 1)^3\Gamma(a)^4},$$

where

$$(5.5) \quad \begin{aligned} Q_*(a, t) = & 2a^4(a + 1)^2\mathcal{R}_a^2 \\ & + 2a^2(a + 1)^2(8a^2 + 3a + 1)\mathcal{R}_a \\ & + 2a(a + 1)(5a + 3). \end{aligned}$$

*Proof.* From (5.1),

$$(5.6) \quad \Delta_2 = S_0 S_2 \text{MD}(M_0, M_2) + S_1^2 \det M_1.$$

Here

$$S_0 = \frac{1}{\Gamma(a)^2}, \quad S_1 = \frac{2}{a\Gamma(a)^2}, \quad S_2 = \frac{2a + 1}{a^2(a + 1)\Gamma(a)^2},$$

and, recalling  $\det M_1 = t - g\beta_1^2$  from (4.11),

$$\text{MD}(M_0, M_2) = g + \frac{g^2}{2a + 1} + \psi_1(a + 2) - 2g\frac{a}{a + 1}.$$

Substituting  $g = t + a^{-2}$  and  $\psi_1(a + 2) = t - (a + 1)^{-2}$  into (5.6) and collecting over the denominator in (5.4) first gives

$$\begin{aligned} Q_*(a, t) = & 2a^4(a + 1)^2t^2 \\ & + 2a^2(8a^4 + 17a^3 + 13a^2 + 5a + 1)t \\ & + 2a(-8a^4 - 10a^3 + a^2 + 7a + 3). \end{aligned}$$

After multiplication by the common denominator in (5.4), the preceding formula is a polynomial identity in  $(a, t)$ . Replacing  $t$  by  $\mathcal{R}_a + (a + 1)^{-1}$  and collecting this polynomial yields the identity (5.5).

Applying (4.2) at  $y = a + 1$  gives  $\mathcal{R}_a = \psi_1(a + 1) - 1/(a + 1) > 1/(2(a + 1)^2) > 0$ . All three coefficients in (5.5) are positive for  $a > 0$ , so  $\Delta_2 > 0$ .  $\square$

*Proof of Theorem 2.2, except fixed- $a$  necessity.* For every  $a > 0$  and  $n \geq 1$ , the endpoint assertion  $\Delta_n^{(1)}(a) > 0$  holds by Lemma 5.2 when  $n \neq 2$  and by Lemma 5.3 when  $n = 2$ . Since  $A$  and  $Z$  have positive coefficients and

$$[\lambda^n]Z_{\Theta_\lambda} = \frac{n}{n!\Gamma(a + n)} > 0 \quad (n \geq 1),$$

every positive-degree coefficient of  $AZZ_{\Theta_\lambda}$  is strictly positive: in degree  $n$ , for example, the product of the constant coefficients of  $A$  and  $Z$  with  $[\lambda^n]Z_{\Theta_\lambda}$  is positive. Therefore

$$\Delta^{(\kappa)} = \Delta^{(1)} + (\kappa - 1)gAZZ_{\Theta_\lambda}$$

shows that all coefficients remain strictly positive for  $\kappa > 1$ .

For the converse, the general  $m = 1$  coefficient gives

$$(5.7) \quad \text{MD}(M_0, M_1^{(\kappa)}) = \frac{\frac{\kappa-1}{2}(ag)^2 + ag - 1}{a^2}.$$

If  $\kappa < 1$ , then  $ag \rightarrow +\infty$  as  $a \downarrow 0$ , so (5.7), and hence  $\Delta_1^{(\kappa)}$ , is negative for all sufficiently small  $a > 0$ . This establishes sufficiency for every  $a > 0$ , and necessity in the form that no  $\kappa < 1$  serves every  $a > 0$ ; necessity at each fixed  $a > 0$  is proved in Corollary 7.2.

Fix  $\kappa \geq 1$ . Since  $\Delta^{(\kappa)}(a, \cdot)$  is entire, each differentiated Maclaurin series converges locally uniformly, so for every  $r \geq 0$  and  $\lambda > 0$ ,

$$\partial_\lambda^r \Delta^{(\kappa)}(a, \lambda) = \sum_{n \geq \max\{1, r\}} \frac{n!}{(n-r)!} \Delta_n^{(\kappa)}(a) \lambda^{n-r} > 0,$$

which makes  $\Delta^{(\kappa)}(a, \cdot)$  strictly absolutely monotone. The leading principal entry satisfies  $A > 0$  for  $\lambda > 0$  by (3.2) and (3.3); since  $\Delta^{(\kappa)} > 0$ , it follows that  $\mathcal{T}_\kappa \succ 0$ .  $\square$

Combining (5.2) with (4.2), which gives  $ag - 1 > (2a)^{-1}$ , yields the explicit lower bound

$$\Delta(a, \lambda) > \Delta_1(a)\lambda = \frac{2(a\psi_1(a) - 1)}{a^3\Gamma(a)^4} \lambda > \frac{\lambda}{a^4\Gamma(a)^4} \quad (a, \lambda > 0).$$

## 6. EXACT REDUCTION FROM ${}_0F_1$ TO THE BESSEL INEQUALITY

Section 2 defines  $G_\nu$ ,  $P_\nu$ ,  $H_\nu$ , and  $D_\nu$  on the Bessel side; the congruence below identifies them with the entries of  $\mathcal{T}$  of (2.5).

The standard identity [19, eq. 10.25.2], [25, §3.7]

$$(6.1) \quad I_{a-1}(2\sqrt{\lambda}) = \lambda^{(a-1)/2} Z(a, \lambda)$$

with  $L = \log Z$ ,  $z = 2\sqrt{\lambda}$ ,  $\nu = a - 1$ , and  $\Theta_z = 2\Theta_\lambda$  gives

$$(6.2) \quad \Theta_z \log I_\nu = \nu + 2L_{\Theta_\lambda},$$

$$P_\nu = 1 + 2L_{a\Theta_\lambda},$$

$$(6.3) \quad G_\nu = -L_{aa},$$

$$H_\nu = 4(L_{\Theta_\lambda} - L_{\Theta_\lambda\Theta_\lambda}).$$

Specializing Remark 2.1 to  $\kappa = 1$ ,

$$(6.4) \quad \frac{A}{Z^2} = -L_{aa}, \quad \frac{B}{Z^2} = 1 + L_{a\Theta_\lambda}, \quad \frac{C}{gZ^2} = \frac{1}{g} + L_{\Theta_\lambda} - L_{\Theta_\lambda\Theta_\lambda}.$$

The Bessel–Schur matrix is therefore the congruence

$$(6.5) \quad \begin{pmatrix} G_\nu & 1 + P_\nu \\ 1 + P_\nu & H_\nu + 4/g \end{pmatrix} = \frac{1}{Z^2} \begin{pmatrix} 1 & 0 \\ 0 & 2/\sqrt{g} \end{pmatrix} \mathcal{T}(a, \lambda) \begin{pmatrix} 1 & 0 \\ 0 & 2/\sqrt{g} \end{pmatrix},$$

whose determinant is

$$(6.6) \quad D_{a-1}(2\sqrt{\lambda}) = \frac{4\Delta(a, \lambda)}{gZ(a, \lambda)^4}.$$

This reduces Theorem 2.4 and Corollary 2.5 to the endpoint  $\kappa = 1$  of Theorem 2.2.

## 7. BESSEL CONSEQUENCES AND SHARPNESS

The endpoint coefficientwise positivity proved in Section 5, transported through the exact reduction of Section 6, gives the sharp Bessel inequality; the boundary limits at  $z \downarrow 0$  and  $z \rightarrow \infty$  fix its optimal constants and complete Theorem 2.2.

*Proof of Theorem 2.4 and Corollary 2.5.* The congruence (6.5) and the endpoint  $\kappa = 1$  of Theorem 2.2, proved in Section 5, give positive definiteness for  $z > 0$ , and (6.6) gives the determinant inequality. As  $z \downarrow 0$ , equivalently  $\lambda \downarrow 0$ , the power series for  $L = \log Z$  gives  $L_{\Theta_\lambda}, L_{a\Theta_\lambda}, L_{\Theta_\lambda\Theta_\lambda} \rightarrow 0$ . Hence (6.2)–(6.3) imply

$$G_\nu \longrightarrow g, \quad 1 + P_\nu \longrightarrow 2, \quad H_\nu + \frac{4}{g} \longrightarrow \frac{4}{g},$$

which is the matrix limit stated in Corollary 2.5.  $\square$

**Lemma 7.1.** *For every  $\nu > -1$  and every  $\kappa \in \mathbb{R}$ ,*

$$(7.1) \quad D_\nu^{(\kappa)}(z) \longrightarrow 2(\kappa - 1) \quad (z \rightarrow \infty),$$

*a limit independent of  $\nu$ .*

*Proof.* Differentiating an asymptotic expansion of  $\log I_\nu$  requires a domain on which it is holomorphic, and  $I_\nu$  has zeros. Fix  $\varphi \in (0, \pi/2)$ , put  $\Sigma_\varphi = \{w : |\arg w| \leq \pi/2 - \varphi\}$ , let  $K \subset \mathbb{C}$  be a closed disc about the given order, and set

$$E_\mu(w) = (2\pi w)^{1/2} e^{-w} I_\mu(w) = 1 - \frac{4\mu^2 - 1}{8w} + O(w^{-2}),$$

the expansion holding uniformly for  $\mu \in K$  and  $w \in \Sigma_\varphi$  [19, eqs. 10.17.1 and 10.40.1, §10.40(iii)]. Hence there is  $W = W(K, \varphi)$  with  $|E_\mu(w) - 1| < 1/2$  whenever  $\mu \in K$ ,  $w \in \Sigma_\varphi$ , and  $|w| \geq W$ . Write  $\Omega$  for the interior of  $K \times (\Sigma_\varphi \cap \{|w| \geq W\})$ . On  $\Omega$  the values of  $E_\mu$  lie in the disc  $|\zeta - 1| < 1/2$ , on which the principal logarithm is holomorphic, so the composition  $\log E_\mu(w)$  is holomorphic in  $(\mu, w)$  and vanishes as  $w \rightarrow \infty$ ; adding  $w - \frac{1}{2} \log(2\pi w)$  with the principal branch of  $\log w$  gives a holomorphic  $\log I_\mu(w)$  on  $\Omega$ , which agrees with the real  $\log I_\nu(z)$  for  $\nu \in K \cap (-1, \infty)$  and  $z > 0$ . Taking logarithms in the expansion of  $E_\mu$  gives

$$(7.2) \quad \log E_\mu(w) = -\frac{4\mu^2 - 1}{8w} + O(w^{-2}),$$

uniformly on  $\Omega$ . Choose  $\varepsilon = \varepsilon(\varphi) \in (0, 1)$  so that  $|w - z| \leq \varepsilon z$  forces  $w \in \Sigma_\varphi$ ; for  $z \geq 2W/(1 - \varepsilon)$  such a disc also lies in  $|w| > W$ . Let  $R(\mu, w)$  denote the  $O(w^{-2})$  remainder in (7.2). Only the given order is needed, so the derivatives are bounded at the centre  $\mu = \nu$  of  $K$ : Cauchy's estimate—on the circle about  $\nu$  of half the radius of  $K$ , and on the disc  $|w - z| \leq \varepsilon z$ , whose radius is proportional to  $|w|$ —gives

$$\partial_\mu^j \partial_w^\ell R(\nu, z) = O(z^{-2-\ell}) \quad (0 \leq j, \ell \leq 2),$$

so each  $w$ -derivative gains a factor  $z^{-1}$  while  $\mu$ -derivatives preserve the order, and (7.2) may be differentiated termwise twice in each variable at  $\mu = \nu$ .

At  $w = z > 0$ , (7.2) therefore gives

$$\log I_\nu(z) = z - \frac{1}{2} \log(2\pi z) - \frac{4\nu^2 - 1}{8z} + O(z^{-2}),$$

and hence  $G_\nu = z^{-1} + O(z^{-2})$ ,  $P_\nu = \nu/z + O(z^{-2})$ , and  $H_\nu^{(\kappa)} = (2\kappa - 1)z - \kappa(1 + 2\nu) + O(z^{-1})$ , which give (7.1).  $\square$

*Proof of Proposition 2.6.* For the least argument-independent correction, replace  $4/\psi_1(\nu + 1)$  in (2.13) by an argument-independent  $R$  and write  $D_{\nu,R}$  for the resulting determinant. Then

$$D_{\nu,R}(z) \longrightarrow \psi_1(\nu + 1)R - 4 \quad (z \downarrow 0),$$

so validity for every  $z > 0$  forces  $R \geq 4/\psi_1(\nu+1)$ . Conversely  $D_{\nu,R} = D_\nu + G_\nu(R - 4/\psi_1(\nu+1)) > 0$  whenever  $R \geq 4/\psi_1(\nu+1)$ , since  $D_\nu > 0$  by Theorem 2.4 and  $G_\nu = A/Z^2 > 0$  for  $\nu > -1$ ,  $z > 0$ .

For the fixed-correction deformation (2.11)–(2.12), the same reduction gives

$$(7.3) \quad D_{a-1}^{(\kappa)}(2\sqrt{\lambda}) = \frac{4\Delta^{(\kappa)}(a, \lambda)}{gZ(a, \lambda)^4}.$$

For each fixed  $\nu > -1$ , necessity of  $\kappa \geq 1$  follows from Lemma 7.1: the limit (7.1) is negative when  $\kappa < 1$ , so  $D_\nu^{(\kappa)}(z) < 0$  for all sufficiently large  $z$ . Sufficiency for  $\kappa \geq 1$  is the part of Theorem 2.2 proved in Section 5, and transfers through (7.3).  $\square$

**Corollary 7.2.** *Fix  $a > 0$  and  $\kappa < 1$ . Then  $\Delta_n^{(\kappa)}(a) < 0$  for some  $n \geq 1$ . Consequently, at every fixed  $a > 0$ , coefficientwise positivity in every positive degree holds if and only if  $\kappa \geq 1$ , which completes Theorem 2.2.*

*Proof of Corollary 7.2, completing Theorem 2.2.* The limit (7.1) of Lemma 7.1 is negative for  $\kappa < 1$ , so  $D_{a-1}^{(\kappa)}(z) < 0$  for all large  $z$ . Hence (7.3) gives  $\Delta^{(\kappa)}(a, \lambda) < 0$  for all large  $\lambda > 0$ . A power series with nonnegative coefficients is nonnegative on  $(0, \infty)$ , so some  $\Delta_n^{(\kappa)}(a)$  is negative. Sufficiency for  $\kappa \geq 1$  is proved in Section 5.  $\square$

**Remark 7.3.** The degree-one argument of Section 5 exhibits failure as  $a \downarrow 0$ . Fix  $a > 0$ ; then no bounded set of degrees detects every  $\kappa < 1$ , since the degree of the first negative coefficient tends to infinity as  $\kappa \uparrow 1$ . In  $\Delta^{(\kappa)} = \Delta^{(1)} + (\kappa - 1)gAZZ_{\Theta_\lambda}$  both series have strictly positive coefficients in every positive degree, so for each  $N$  the finitely many inequalities  $\Delta_n^{(\kappa)}(a) > 0$ ,  $1 \leq n \leq N$ , hold once  $\kappa < 1$  is close enough to 1.

## 8. CONTINUATION BEYOND THE POSITIVE-SERIES DOMAIN

The behavior changes qualitatively once the order crosses the endpoint  $\nu = -1$  of the positive-series domain. On the open set

$$\mathcal{U} = \{(\nu, z) \in (\mathbb{R} \setminus \{-1, -2, \dots\}) \times (0, \infty) : I_\nu(z) \neq 0\},$$

define  $G_\nu$ ,  $P_\nu$ ,  $H_\nu$ , and  $D_\nu$  by (2.7)–(2.10) with  $\log I_\nu$  replaced by  $\log |I_\nu(z)|$ ; the order derivatives are taken on  $\mathcal{U}$ , and for each fixed  $\nu$  the resulting formulas restrict to any connected component of  $\{z > 0 : I_\nu(z) \neq 0\}$ . Where  $I_\nu > 0$  this agrees with the ordinary logarithm.

For every real  $a \notin \{0, -1, -2, \dots\}$ , the partial-fraction expansion

$$(8.1) \quad \psi_1(a) = \sum_{r=0}^{\infty} \frac{1}{(a+r)^2}$$

holds and shows that  $g = \psi_1(a) > 0$  [20, eq. 5.15.1].

**Lemma 8.1.** *The series (2.1) and all of its fixed-order parameter and Euler derivatives converge locally uniformly on  $\mathbb{C}^2$ , so  $Z$  and those derivatives are entire in  $(a, \lambda)$ . For each fixed coefficient index, the scalar coefficient identities in (3.2)–(3.7), and every resulting finite-convolution formula for a coefficient of  $\Delta$ , extend meromorphically in  $a$ . A formula carrying a product  $S_m c_m^{(\kappa)}$  is to be read as that complete product, whose apparent pole at  $2a + 2m - 3 = 0$  is cancelled by the zero of  $S_m$  there and is therefore removable. This assertion concerns the scalar entries and  $\Delta$ ; it does not require a single-valued meromorphic choice of  $\sqrt{\psi_1(a)}$  on the complex plane. If  $a \in \mathbb{R} \setminus \{0, -1, -2, \dots\}$ ,  $\lambda > 0$ , and  $Z(a, \lambda) \neq 0$ , then (6.4), (6.5), and (6.6) remain valid, with the positive real square root of  $\psi_1(a)$  and with logarithmic derivatives interpreted through  $\log |Z|$  and  $\log |I_{a-1}|$  on the relevant nonvanishing component.*

*Proof.* Each summand of (2.1) is entire in  $a$ . Uniform Stirling estimates for  $a$  in compact sets, together with Cauchy estimates for the derivatives of  $1/\Gamma$ , give locally uniform convergence of the series and of every fixed-order parameter and Euler derivative on compact subsets of  $\mathbb{C}^2$ ; hence  $Z$  and these derivatives are entire in  $(a, \lambda)$ . For each fixed coefficient index, both sides

of each scalar coefficient identity are meromorphic functions of the single variable  $a$  and agree on  $a > 0$ ; since  $a > 0$  has a limit point, the one-variable identity theorem shows they agree throughout their common domain, the complement in  $\mathbb{C}$  of the discrete pole set. Each coefficient of  $\Delta = AC - gB^2$  is a finite convolution of such scalar coefficient functions and is therefore meromorphic as well.

On the real nonpole set, (8.1) gives  $g = \psi_1(a) > 0$ , so the positive real square root used in the matrix formulas is well defined. Let

$$\mathcal{V} = \{(a, \lambda) : a \in \mathbb{R} \setminus \{0, -1, -2, \dots\}, \lambda > 0, Z(a, \lambda) \neq 0\}.$$

On each connected component of  $\mathcal{V}$ ,  $Z$  has constant sign, so  $\partial_a \log |Z| = Z_a/Z$  and the corresponding mixed and second derivative identities hold there. Finally, (6.1) gives  $\log |I_{a-1}(2\sqrt{\lambda})| = \frac{a-1}{2} \log \lambda + \log |Z(a, \lambda)|$  where  $Z \neq 0$ , leaving the derivations of (6.4), (6.5), and (6.6) unchanged on that component.  $\square$

Away from the poles the leading small- $z$  term keeps its positive sign. For  $a = \nu + 1 \in \mathbb{R} \setminus \{0, -1, -2, \dots\}$  the ascending series gives  $I_\nu(z) = (z/2)^\nu \Gamma(\nu + 1)^{-1} (1 + O(z^2))$ , so  $I_\nu$  has constant sign near  $z = 0$ ; by Lemma 8.1, (5.2) and (6.6) with  $Z(a, 0) = 1/\Gamma(a)$  and  $\lambda = z^2/4$  give

$$(8.2) \quad D_\nu(z) = \frac{2(a\psi_1(a) - 1)}{a^3\psi_1(a)} z^2 + O(z^4) \quad (z \downarrow 0),$$

whose coefficient is positive for every such  $a$ : for  $a > 0$  by (4.2), and for negative nonintegral  $a$  because (8.1) makes  $a\psi_1(a) - 1$  and  $a^3\psi_1(a)$  both negative. Coefficientwise positivity of  $\Delta$  nonetheless fails already on the next interval where  $1/\Gamma(a) > 0$ : for  $-3 < \nu < -2$  one has  $a \in (-2, -1)$  and  $\Gamma(a) > 0$ .

**Proposition 8.2** (Degree-two obstruction beyond the positive-series domain). *If  $-2 < a < -1$ , equivalently  $-3 < \nu < -2$ , then*

$$(8.3) \quad [\lambda^2]\Delta(a, \lambda) = \Delta_2(a) < 0.$$

*Thus the coefficientwise positivity theorem for  $\Delta$  cannot be extended to that interval.*

*Proof.* By Lemma 8.1, the identity (5.4)–(5.5) persists on  $-2 < a < -1$ . Set  $b = -a \in (1, 2)$ , so that  $a + 1 \in (-1, 0)$ . The denominator in (5.4) is then negative:  $a^6 > 0$  and  $\Gamma(a)^4 > 0$ , while  $(a + 1)^3 < 0$ . It therefore suffices to show  $Q_*(a, t) > 0$ . By (8.1),  $t = \psi_1(a + 1) > 0$  and

$$\mathcal{R}_a = t - \frac{1}{a+1} > \frac{1}{b-1}.$$

The first term in (5.5) is positive. The second and third terms satisfy

$$\begin{aligned} & 2b^2(b-1)^2(8b^2 - 3b + 1)\mathcal{R}_a - 2b(b-1)(5b-3) \\ & > 2b(b-1) [8b^3 - 3b^2 - 4b + 3] > 0. \end{aligned}$$

Indeed, the polynomial in brackets equals 4 at  $b = 1$  and has derivative  $24b^2 - 6b - 4 > 0$  for  $b \geq 1$ . Hence  $Q_*(a, t) > 0$ , proving (8.3).  $\square$

A complete classification beyond  $\nu > -1$  appears to require separate treatment.

## 9. COMPARISON WITH RELATED POSITIVITY RESULTS

The scalar quantity  $A = Z_a^2 - ZZ_{aa}$  is the infinitesimal symmetric-shift Turánian of the reciprocal-gamma series. Indeed,  $Z(a + \delta, \lambda)Z(a - \delta, \lambda) - Z(a, \lambda)^2 = -A(a, \lambda)\delta^2 + O(\delta^4)$ . The scalar coefficient sign of  $A$  is the infinitesimal case of the coefficientwise reciprocal-gamma Turán inequalities of Kalmykov and Karp [11, Thm. 3.1 and Ex. 1] (the integer-shift form in [14, Lem. 1 and Thm. 3]), whose gamma-ratio results similarly treat scalar two-function Turánians under parameter shifts [12, 13, 15]. The coefficientwise thresholds of Karp and Zhang [16, §3] are set by the multiplicity of Pochhammer symbols in such scalar series; the threshold here is instead the deformation parameter  $\kappa$  of a coupled matrix determinant.

The additional quantities  $B$  and  $C$  are not scalar parameter-shift Turánians:  $B$  contains a mixed parameter–Euler derivative, while  $C$  contains the sharp argument-direction expression and the trigamma normalization that becomes the least argument-independent correction under the Bessel congruence. The result proved here is the coefficientwise positivity of the coupled determinant  $AC - \psi_1(a)B^2$ .

Classical Bessel Turán inequalities compare neighboring orders, for example  $I_\nu^2 - I_{\nu-1}I_{\nu+1}$ , or control ratios and logarithmic derivatives [1, 2, 4, 24]; series representations of the modified-Bessel Turánian [17, Thms. 1 and 5], positivity of Turán determinants for symmetric orthogonal polynomials [7, Thm. 1 and Cor. 1], and higher-order Turán and Hankel determinants in the order [3, §§2.2, 3] have also been studied, as has the absolute monotonicity in the argument of Hankel determinants with special-function entries [10, Thm. 2.3]. In contrast, (2.13) couples second curvature in the order with first mixed order–argument curvature and an argument-direction expression. Derivatives of  $I_\nu$  with respect to the order have been studied in their own right, with closed forms and integral representations for  $\partial_\nu I_\nu$  [21, §10.38], [9] and a closed form for  $\partial_\nu^2 I_\nu$  at integer orders [5, eq. (7.1)]. For  $J_\nu$  and  $Y_\nu$  the corresponding integral representations, with uniform Airy asymptotics for large order, are those of [8, Props. 2.1 and 2.2, Thm. 4.2]; the modified-function representations of [9] follow a similar derivation. The quantity  $P_\nu$  couples a first derivative in  $\nu$  to the argument, whereas  $G_\nu$  is second order in  $\nu$  and is required for every  $\nu > -1$ , not only at integer orders. Positivity of the leading entry  $G_\nu$  alone is the strict log-concavity of  $\nu \mapsto I_\nu(z)$  on  $(-1, \infty)$ , which is known [11, Ex. 1]; a coefficient argument gives it on the same range in the two-sided shifted form  $1 < I_\nu^2/(I_{\nu+\delta}I_{\nu-\delta}) < \Gamma(\nu+\delta+1)\Gamma(\nu-\delta+1)/\Gamma(\nu+1)^2$  for  $0 < \delta < \nu+1$  [18, Thm. 3]. These results concern individual order derivatives or scalar order log-concavity; neither the coupled determinant nor the pointwise inequality of Theorem 2.4, with its argument-independent correction  $4/\psi_1(\nu+1)$ , appears to have an antecedent among them.

A recent preprint of Salazar establishes that the kernel  $(z, \nu) \mapsto I_\nu(z)$  is strictly totally positive for  $z > 0$  and  $\nu \geq 0$  [22], extending integer-order results of Buchstaber and Glutsyuk [6]; a companion preprint builds all-order Hankel and Turán determinant hierarchies for special functions from positive moment representations [23]. Their determinant hierarchies are, respectively, of evaluation-minor and Hankel-moment type; none of these papers states or proves the coefficientwise mixed-derivative determinant studied here. The signs of all total-positivity minors of  $(z, \nu) \mapsto I_\nu(z)$ , including their confluent limits, are preserved under positive row and column rescalings of the kernel, since a positive factor acts on a confluent jet matrix by a triangular transformation with positive diagonal and multiplies each minor by a positive number. The Schur determinant of (2.13) is not preserved. To make this explicit, write  $F(\nu, y) = \log I_\nu(e^y)$ , so that  $\partial_y = \Theta_z$  under  $z = e^y$  and (2.7)–(2.9) read

$$G_\nu = -F_{\nu\nu}, \quad P_\nu = F_{\nu y}, \quad H_\nu = 2(F_y - \nu) - F_{yy}.$$

Replacing  $I_\nu(e^y)$  by  $e^{cy}I_\nu(e^y)$  gives  $\tilde{F} = F + cy$ , with

$$\tilde{G}_\nu = -\tilde{F}_{\nu\nu} = G_\nu, \quad \tilde{P}_\nu = \tilde{F}_{\nu y} = P_\nu, \quad \tilde{H}_\nu = 2(\tilde{F}_y - \nu) - \tilde{F}_{yy} = H_\nu + 2c.$$

By (2.10) the rescaled kernel therefore has

$$\tilde{D}_\nu(z) = D_\nu(z) + 2cG_\nu(z).$$

Here  $G_\nu = A/Z^2 > 0$ , so at any fixed  $z_0 > 0$  every  $c < -D_\nu(z_0)/(2G_\nu(z_0))$  gives  $\tilde{D}_\nu(z_0) < 0$ . The sign of  $D_\nu$  is therefore not invariant under positive rescaling of the kernel, so (2.13) is not a rescaling-covariant confluent total-positivity minor.

## 10. FURTHER QUESTIONS

**Question 10.1.** The series  $Z$  is the  $\rho = 1$  member of a Wright-type family. For  $\rho > 0$ , let

$$Z_\rho(a, \lambda) = \sum_{k \geq 0} \frac{\lambda^k}{k! \Gamma(a + \rho k)},$$

and, for  $\kappa \in \mathbb{R}$ , define

$$\begin{aligned} \frac{A_\rho}{Z_\rho^2} &= -\partial_a^2 \log Z_\rho, & \frac{B_\rho}{Z_\rho^2} &= 1 + \rho \partial_a \Theta_\lambda \log Z_\rho, \\ \frac{C_{\rho,\kappa}}{\psi_1(a)Z_\rho^2} &= \frac{1}{\psi_1(a)} + \kappa \rho \Theta_\lambda \log Z_\rho - \rho^2 \Theta_\lambda^2 \log Z_\rho, & \Delta_\rho^{(\kappa)} &= A_\rho C_{\rho,\kappa} - \psi_1(a) B_\rho^2. \end{aligned}$$

For which pairs  $(\rho, \kappa)$  does  $\Delta_\rho^{(\kappa)}$  have strictly positive Maclaurin coefficients in every positive degree for all  $a > 0$ ; and for each  $\rho > 0$ , what is the sharp threshold in  $\kappa$ ? At  $\rho = 1$  the present theorem gives exactly  $\kappa \geq 1$ , and its proof makes essential use of the exact  $\rho = 1$  Chu–Vandermonde convolution.

**Question 10.2.** Is there a canonical normalization, bordered kernel, or nonlocal minor associated with  $(z, \nu) \mapsto I_\nu(z)$  whose positivity is equivalent to the Bessel–Schur inequality of Theorem 2.4?

The rescaling computation of Section 9 shows that any such derivation must incorporate additional Bessel-specific normalization or nonlocal information. A related question is whether the coefficientwise positivity of  $AC - \psi_1(a)B^2$  follows from an appropriately strengthened form of scalar reciprocal-gamma log-concavity.

#### DECLARATIONS

**Funding.** This research received no external funding.

**Competing interests.** The author has no relevant interests to disclose.

**Data availability.** No datasets were generated or analysed.

**Computational verification.** The proofs do not depend on numerical computation. Scripts used for verification and figures are available at <https://github.com/johnnyshields/papers>.

**Generative AI use.** During the preparation of this work, the author used OpenAI’s ChatGPT (GPT-5.6 Sol) and Anthropic’s Claude (Opus 5, Opus 4.8) to assist with manuscript drafting and the development of auxiliary verification code. After using these tools, the author reviewed and edited as needed. The author takes full responsibility for the content of the work.

#### REFERENCES

- [1] Á. Baricz, *Functional inequalities involving Bessel and modified Bessel functions of the first kind*, Expo. Math. **26** (2008), 279–293, DOI: 10.1016/j.exmath.2008.01.001.
- [2] Á. Baricz, *Turán type inequalities for modified Bessel functions*, Bull. Aust. Math. Soc. **82** (2010), 254–264, DOI: 10.1017/S000497271000002X.
- [3] Á. Baricz and T. K. Pogány, *Turán determinants of Bessel functions*, Forum Math. **26** (2014), 295–322, DOI: 10.1515/form.2011.160.
- [4] Á. Baricz and S. Ponnusamy, *On Turán type inequalities for modified Bessel functions*, Proc. Amer. Math. Soc. **141** (2013), 523–532, DOI: 10.1090/S0002-9939-2012-11325-5.
- [5] Yu. A. Brychkov, *Higher derivatives of the Bessel functions with respect to the order*, Integral Transforms Spec. Funct. **27** (2016), 566–577, DOI: 10.1080/10652469.2016.1164156.
- [6] V. M. Buchstaber and A. A. Glutsyuk, *Total positivity, Grassmannian and modified Bessel functions*, in *Functional Analysis and Geometry: Selim Grigorievich Krein Centennial*, Contemp. Math. **733**, Amer. Math. Soc., 2019, 97–107, DOI: 10.1090/conm/733/14736.
- [7] J. Bustoz and M. E. H. Ismail, *Turán inequalities for symmetric orthogonal polynomials*, Int. J. Math. Math. Sci. **20** (1997), 1–7, DOI: 10.1155/S016117129700001X.
- [8] T. M. Dunster, *On the order derivatives of Bessel functions*, Constr. Approx. **46** (2017), 47–68, DOI: 10.1007/s00365-016-9355-1.
- [9] J. L. González-Santander, *Closed-form expressions for derivatives of Bessel functions with respect to the order*, J. Math. Anal. Appl. **466** (2018), 1060–1081, DOI: 10.1016/j.jmaa.2018.06.043.

- [10] M. E. H. Ismail and A. Laforgia, *Monotonicity properties of determinants of special functions*, Constr. Approx. **26** (2007), 1–9, DOI: 10.1007/s00365-005-0627-4.
- [11] S. I. Kalmykov and D. B. Karp, *Log-concavity for series in reciprocal gamma functions and applications*, Integral Transforms Spec. Funct. **24** (2013), 859–872, DOI: 10.1080/10652469.2013.764874.
- [12] S. I. Kalmykov and D. B. Karp, *Log-convexity and log-concavity for series in gamma ratios and applications*, J. Math. Anal. Appl. **406** (2013), 400–418, DOI: 10.1016/j.jmaa.2013.04.061.
- [13] S. I. Kalmykov and D. B. Karp, *Logarithmic concavity of series in gamma ratios*, Russian Math. **58** (2014), 63–68, DOI: 10.3103/S1066369X14060073.
- [14] S. I. Kalmykov and D. B. Karp, *Log-concavity and Turán-type inequalities for the generalized hypergeometric function*, Anal. Math. **43** (2017), 567–580, DOI: 10.1007/s10476-017-0503-z.
- [15] D. Karp and S. M. Sitnik, *Log-convexity and log-concavity of hypergeometric-like functions*, J. Math. Anal. Appl. **364** (2010), 384–394, DOI: 10.1016/j.jmaa.2009.10.057.
- [16] D. Karp and Y. Zhang, *Log-concavity and log-convexity of series containing multiple Pochhammer symbols*, Fract. Calc. Appl. Anal. **27** (2024), 458–486, DOI: 10.1007/s13540-023-00238-0.
- [17] I. Mező and Á. Baricz, *Properties of the Turánian of modified Bessel functions*, Math. Inequal. Appl. **20** (2017), 991–1001, DOI: 10.7153/mia-2017-20-63.
- [18] T. Nanthanasub, B. Novaprateep and N. Wichailukkana, *The logarithmic concavity of modified Bessel functions of the first kind and its related functions*, Adv. Difference Equ. **2019** (2019), art. 379, 14 pp., DOI: 10.1186/s13662-019-2309-8.
- [19] NIST Digital Library of Mathematical Functions, *Chapter 10: Bessel Functions*, <https://dlmf.nist.gov/10>, accessed 22 July 2026.
- [20] NIST Digital Library of Mathematical Functions, *Chapter 5: Gamma Function*, <https://dlmf.nist.gov/5>, accessed 19 July 2026.
- [21] NIST Digital Library of Mathematical Functions, *Section 10.38: Derivatives with Respect to Order*, <https://dlmf.nist.gov/10.38>, accessed 22 July 2026.
- [22] D. S. P. Salazar, *Strict total positivity from spectral Darboux and Toeplitz smoothing mechanisms*, arXiv:2607.02778, 2026, DOI: 10.48550/arXiv.2607.02778.
- [23] D. S. P. Salazar, *Order-moment transport and Hankel determinants in special-function inequalities*, arXiv:2606.31647, 2026, DOI: 10.48550/arXiv.2606.31647.
- [24] J. Segura, *Bounds for ratios of modified Bessel functions and associated Turán-type inequalities*, J. Math. Anal. Appl. **374** (2011), 516–528, DOI: 10.1016/j.jmaa.2010.09.030.
- [25] G. N. Watson, *A Treatise on the Theory of Bessel Functions*, 2nd ed., Cambridge University Press, 1944.